

Mark schemes

Q1.

$$m^2 + 2m + 10 = 0$$

PI

M1

$$m = -1 \pm 3i$$

A1

Complementary function is $(y =) e^{-x} (A \cos 3x + B \sin 3x)$

*OE Ft on incorrect **complex value** of m*

A1F

Particular integral: try $y = ke^x$

M1

$$k + 2k + 10k = 26 \Rightarrow k = 2$$

A1

$$(GS \ y =) e^{-x} (A \cos 3x + B \sin 3x) + 2e^x$$

c's CF + c's non-zero PI but must have 2 arb consts

B1F

$$x = 0, y = 5 \Rightarrow 5 = A + 2 \text{ so } A = 3$$

ft c's k ie $A = 5 - k, k \neq 0$

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$$\frac{dy}{dx} =$$

*Attempt to differentiate c's **GS***

$$e^{-x} (-3A \sin 3x + 3B \cos 3x - A \cos 3x - B \sin 3x) + 2e^x$$

(ie CF + PI)

$$11 = 3B - A + 2 \quad (B = 4)$$

A1

$$y = e^{-x} (3 \cos 3x + 4 \sin 3x) + 2e^x$$

CSO

A1

[10]

Q2.

Aux. eqn. $m^2 + 4 = 0 \Rightarrow m = \pm 2i$

B1

CF is $A \cos 2x + B \sin 2x$

OE. If m is real give M0

M1

ft on incorrect complex value for m

A1F

PI: Try $ax^2 + b + c \sin x$

Award even if extra terms, provided the relevant coefficients are shown to be zero.

M1M1

$$2a - c \sin x + 4ax^2 + 4b + 4c \sin x = 8x^2 + 9 \sin x$$

$$a = 2, b = -1,$$

Dep on relevant M mark

A1

$$c = 3$$

Dep on relevant M mark

A1

$$(y =) A \cos 2x + B \sin 2x + 2x^2 - 1 + 3 \sin x$$

Their CF + their PI. Must be exactly two arbitrary constants

B1F

[8]

Q3.

(a) $-4k \sin 2x + k \sin 2x = \sin 2x$

Substituting into the differential equation

M1A1

$$k = -\frac{1}{3}$$

Accept correct PI

A1

3

(b) (Aux. eqn $m^2 + 1 = 0$) $m = \pm i$

PI

B1

CF: $A \cos x + B \sin x$

M0 if m is real

M1

*OE Ft on incorrect complex values for m
For the A1F do not accept if left in the
form $Ae^{ix} + Be^{-ix}$*

A1F

(GS: $y =$) $A \cos x + B \sin x - \frac{1}{3} \sin 2x$
c's CF + c's PI but must have 2 constants

B1F

4

[7]

Q4.

(a) $-k \sin x + 2k \cos x + 5k \sin x = 8 \sin x + 4 \cos x$
Differentiation and subst. into DE

M1A1

$k = 2$

A1

3

(b) Auxl eqn $m^2 + 2m + 5 = 0$
 $m = \frac{-2 \pm \sqrt{4 - 20}}{2}$

Formula or completing sq. PI

M1

$m = -1 \pm 2i$

A1

CF: $\{y_C\} = e^{-x}(A \sin 2x + B \cos 2x)$

ft provided m is not real

A1F

GS $\{y\} = e^{-x}(A \sin 2x + B \cos 2x) + k \sin x$

ft on CF + PI; must have 2 arb consts

B1F

When $x = 0, y = 1 \Rightarrow B = 1$

B1F

$$\frac{dy}{dx} = -e^{-x} (A \sin 2x + B \cos 2x) + e^{-x} (2A \cos 2x - 2B \sin 2x) + k \cos x$$

Product rule

M1

When $x = 0, \frac{dy}{dx} = 4 \Rightarrow 4 = -B + 2A + k$
PI

A1

$$\Rightarrow A = \frac{3}{2}$$

$$y = e^{-x} \left(\frac{3}{2} \sin 2x + \cos 2x \right) + 2 \sin x$$

CSO

A1

8

[11]

Q5.

(a) $(m + 2)^2 = -1$

Completing sq or formula

M1

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$$m = -2 \pm i$$

A1

CF is $e^{-2x} (A \cos x + B \sin x)$

If m is real give M0

M1

{or $e^{-x} A \cos (x + B)$
but not $Ae^{(-1+i)x} + Be^{(-1-i)x}$ }

Ft on wrong a's and b's but roots must be complex

A1ft

PI try $y = p \Rightarrow 5p = 5$ PI is $y = 1$

B1

$$\text{GS } y = e^{-2x} (A \cos x + B \sin x) + 1$$

Their CF + their PI with two arbitrary constants.

B1ft

6

(b) $x = 0, y = 2 \Rightarrow A = 1$

Provided previous B1ft awarded

B1ft

$$y'(x) = -2e^{-2x} (A \cos x + B \sin x) + e^{-2x} (-A \sin x + B \cos x)$$

Product rule used

M1A1ft

$$y'(0) = 3 \Rightarrow 3 = -2A + B \Rightarrow B = 5$$

Ft on one slip

A1ft

$$y = e^{-2x} (\cos x + 5 \sin x) + 1$$

4

[10]

Q6.

(a) Aux eqn: $m^2 - 2m - 3 = 0$

M1

$$m = -1, 3$$

PI

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A1

$$\text{CF } (y_C) = Ae^{3x} + Be^{-x}$$

M1

$$\text{Try } (y_{PI}) = a e^{-2x} \quad (+ b)$$

M1

$$\frac{dy}{dx} = -2ae^{-2x}$$

A1

$$\frac{d^2y}{dx^2} = 4ae^{-2x}$$

A1

Substitute into DE gives

$$4ae^{-2x} + 4ae^{-2x} - 3ae^{-2x} - 3b = 10e^{-2x} - 9$$

M1

$$\Rightarrow a = 2$$

A1

$$b = 3$$

B1

$$(y_{GS} =) Ae^{3x} + Be^{-x} + 2e^{-2x} + 3$$

(c's CF+c's PI) with 2 arbitrary constants

B1F

10

(b) $x = 0, y = 7 \Rightarrow 7 = A + B + 2 + 3$

Only fit if exponentials in GS and two arbitrary constants remain

B1F

$$\frac{dy}{dx} = 3Ae^{3x} - Be^{-x} - 4e^{-2x}$$

As $x \rightarrow \infty, e^{-kx} \rightarrow 0, \frac{dy}{dx} \rightarrow 0$ so $A = 0$

B1

When $A = 0, 5 = 0 + B + 3 \Rightarrow B = 2$

Must be using 'A' = 0

B1F

$$y = 2e^{-x} + 2e^{-2x} + 3$$

CSO

A1

4

[14]

Q7.

Auxl. eqn $m^2 - 4m + 3 = 0$

PI

M1

$$m = 3 \text{ and } 1$$

PI

A1

CF is $A e^{3x} + B e^x$

A1F

PI Try $y = a + b \sin x + c \cos x$

Condone 'a' missing here

M1

$$y'(x) = b \cos x - c \sin x$$

A1

$$y''(x) = -b \sin x - c \cos x$$

ft can be consistent sign error(s)

A1F

Substitute into DE gives

M1

$$a = 2$$

B1

$$4c + 2b = 5 \text{ and } 2c - 4b = 0$$

A1

$$b = 0.5,$$

ft a slip

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$$c = 1$$

ft a slip

A1F

$$\text{GS: } y = A e^{3x} + B e^x + 2 + 0.5 \sin x + \cos x$$

*y = candidate's CF and candidate's PI
(must have exactly two arbitrary constants)*

B1F

12

[12]

Q8.

(a) $(m + 1)^2 = -1$

Completing sq or formula

M1

$$m = -1 \pm i$$

A1

2

- (b) (i) CF is $e^{-x}(A \cos x + B \sin x)$
If m is real give M0

M1

{or $e^{-x}A \cos(x + B)$ **but not** $Ae^{(-1+i)x} + Be^{(-1-i)x}$ }
ft On wrong a 's and b 's but roots must be complex.

A1 ✓

{P.Int.} try $y = px + q$

OE

$$2p + 2(px + q) = 4x$$

$$p = 2, \quad q = -2$$

On one slip

M1

A1

A1 ✓

GS $y = e^{-x}(A \cos x + B \sin x) + 2x - 2$

Their CF + their PI with two arbitrary constants.

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6

- (ii) $x = 0, y = 1 \Rightarrow A = 3$
Provided an M1 gained in (b)(i)

B1 ✓

$$y'(x) = -e^{-x}(A \cos x + B \sin x) +$$

Product rule used

M1

$$+ e^{-x}(-A \sin x + B \cos x) + 2$$

A1 ✓

$$y'(0) = 2 \Rightarrow 2 = -A + B + 2 \Rightarrow B = 3$$

Slips

A1 ✓

$$y = 3e^{-x}(\cos x + \sin x) + 2x - 2$$

4

[12]

Q9.

(a) $y = 2x + \sin 2x \Rightarrow y' = 2 + 2\cos 2x$
 $\Rightarrow y'' = -4\sin 2x$

Need to attempt both y' and y''

M1 A1

$$-4\sin 2x - 5(2 + 2\cos 2x) + 4(2x + \sin 2x) =$$

$$8x - 10 - 10\cos 2x$$

CSO AG Substitute. and confirm correct

A1

3

(b) Auxiliary equation $m^2 - 5m + 4 = 0$

M1

$$m = 4 \text{ and } 1$$

A1

$$\text{CF: } Ae^{4x} + Be^x$$

M1

$$\text{GS: } y = Ae^{4x} + Be^x + 2x + \sin 2x$$

Their CF + $2x + \sin 2x$

B1ft

4

(c) $x = 0, y = 2 \Rightarrow 2 = A + B$

Only ft if exponentials in GS

B1ft

$$x = 0, y' = 0 \Rightarrow 0 = 4A + B + 4$$

Only ft if exponentials in GS and differentiated four terms at least

B1ft

Solving the simultaneous equations

M1

gives $A = -2$ and $B = 4$

A1

4

$$y = -2e^{4x} + 4e^x + 2x + \sin 2x$$

[11]

Q10.

(a) Substituting $z = Ae^{nt}$,

$$2n^2 + 6n + 9 = 0$$

M1

$$n = \left(-6 \pm \frac{\sqrt{36 - 72}}{4} \right)$$

$$n = -\frac{3}{2} \pm \frac{3}{2}i$$

A1

$$\therefore x = e^{-\frac{3}{2}t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right)$$

M1 A1ft

When $t = 0$, $x = a$; $a = A$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{3}{2}e^{-\frac{3}{2}t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right) + \frac{3}{2}e^{-\frac{3}{2}t} \left(-A \sin \frac{3}{2}t + B \cos \frac{3}{2}t \right)$$

Use of product formula

M1 A1

$$\text{When } t = 0, \frac{dx}{dt} = 0; 0 = -\frac{3}{2}A + \frac{3}{2}B$$

M1

$$B = A$$

$$\therefore x = Ae^{-\frac{3}{2}t} \left(\cos \frac{3}{2}t + \sin \frac{3}{2}t \right)$$

A1

9

- (b) Damping motion is light damping
And one reason

B1

since the particle oscillates, with its amplitude reducing to zero

B1

2

Or

$e^{-kt} \times \text{trig. term}$

Or

Sketch from graphics calculator

Full reason

[11]



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Examiner reports

Q1.

The majority of candidates scored full marks for solving this second order differential equation, subject to the given boundary conditions, correctly. Other than arithmetic and algebraic slips, the main source of misunderstanding was to give the complementary function as $Ae^{-x}(\cos 3x + \sin 3x)$, which lead to a general solution of a second order differential equation with only one arbitrary constant.

Q2.

This unstructured question was answered well by many of the candidates. However, a significant minority of candidates did not even write down the correct form of the auxiliary equation, or they solved it incorrectly to give real rather than imaginary values. When finding the particular integral, a large number of candidates considered more terms than they needed to, and they did not always go on to show that the relevant coefficients of the extra terms were zero. The majority of candidates knew that they had to find a complementary function and a particular integral and almost all scored the final mark for combining the two to give the general solution.

Q3.

This was another very good source of marks for candidates, with many fully correct solutions presented. In part (a) some candidates decided to ignore the given form of the particular integral and worked with $p\cos 2x + q\sin 2x$. Such an approach was not penalised

by examiners provided the candidate showed that both $p = 0$ and $q = -\frac{1}{3}$. The vast majority of candidates showed that they knew the methods to solve the second order differential equation but errors in forming and solving the auxiliary equation were sources of loss of marks. Real solutions from the auxiliary equation were more heavily penalised than other errors.

Q4.

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In part (a), many candidates decided to ignore the given form of the particular integral and worked with $a \cos x + b \sin x$. Such an approach was not penalised by examiners provided the candidate showed that both $a = 0$ and $b = 2$. The vast majority of candidates showed that they knew the methods required to solve the second order differential equation, but arithmetical errors in solving the auxiliary equation $m^2 + 2m + 5 = 0$ or in

applying the boundary condition $\frac{dy}{dx} = 4$ when $x = 0$ or the wrong differentiation of $\cos 2x$ were sources of loss of marks as well as the more serious error of applying the boundary conditions to the complementary function before adding on the particular integral.

Q5.

This question, which required candidates to solve a second order differential equation, was generally a good source of marks. It was disappointing to see some candidates trying to solve the auxiliary equation, $m^2 + 4m + 5 = 0$, by factorisation. They obtained real solutions and this error was penalised heavily. Better candidates were able to write down the correct complementary function and find the particular integral but some wasted valuable time by starting with $y_p = ax^2 + bx + c$ and showing that both a and b were zero.

Candidates who were able to find the correct general solution in part (a) usually went on to apply the given boundary conditions correctly in their answers to part (b).

Q6.

In part (a), many candidates scored high marks for finding the general solution of the second order differential equation, with the most common slip being a wrong expansion of brackets, which led to -9 instead of -3 for the constant part of the particular integral. A more serious but less common error was to look for a particular integral of the form $axe^{-2x} + b$. Part (b) proved, as expected, to be more of a challenge to candidates, with many unable to deal with the boundary condition expressed as a limit. This was the first time that this had appeared on an examination paper for this unit and, in general, only the better candidates could handle it.

Q7.

This question, on finding the general solution of a second order differential equation, was a good source of marks for many candidates. Some candidates failed to write their final answer in the form $y = f(x)$, but a more common error was not finding/stating “2” as part of the particular integral.

Q8.

The majority of candidates gained at least ten of the twelve available marks for this opening question which tested the solution of a second order differential equation. Part (a) caused few problems although there were a few who had difficulty simplifying

$$\frac{\sqrt{4-8}}{2}$$

correctly. The most common method error in the question occurred in part(b)(i). These candidates considered a particular integral of the form ‘ ax ’ instead of the correct form ‘ $ax + b$ ’. In general, candidates were able to apply the given boundary conditions in a correct manner.

Q9.

Most candidates gained very high marks for this opening question which tested the solution of a second order differential equation. In part (a) it was acceptable to differentiate the given result and substitute into the given differential equation. Although sign errors were seen, the vast majority who used this approach scored all three marks. Those who took a general form for the particular integral sometimes failed to find all the unknown constants.

Part (b) was generally answered correctly although some gave the final answer as $y = Ae^{4x} + Be^{2x} + 8x - 10 - 10\cos 2x$. Except for some careless slips in using $y'(0) = 0$ the final part of the question was answered well.

Q10.

Part (a) was completed well by most candidates. However the use of the formula to solve the quadratic equation $2n^2 + 6n + 9 = 0$ was completed incorrectly by a number of candidates. Most appreciated that the damping was light, although the reason given for this was often imprecise.